

Test_Strategy_Case_Standard

Technical Standard

保险公司 技术开发部
内部文档 · 仅供授权人员查阅

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Test Strategy & Case Standard

> Reference: Google Testing Certification, Alibaba Testing Standards, ThoughtWorks Test Pyramid, ISTQB Standards

1. Overview

1.1 Purpose

Standardize test strategy, case design, and execution processes for insurance core systems, ensuring sufficient test coverage and measurable quality, with special focus on accounting testing and actuarial interface verification.

1.2 Scope

This standard applies to functional testing, integration testing, accounting testing, and actuarial interface verification for all products/projects within the Technology Development Department.

2. Test Pyramid & Strategy

2.1 Test Layer Strategy

Layer	Proportion	Speed	Responsibility	Tools
Unit Test (UT)	60%	Milliseconds	Verify function/method logic	JUnit / pytest / Go test
Integration Test (IT)	25%	Seconds	Verify module interaction, DB operations, API calls	Testcontainers / Mockito
End-to-End Test (E2E)	10%	Minutes	Verify complete business workflows	Selenium / Cypress
Manual Exploratory	5%	N/A	Edge cases, complex business logic	Manual

2.2 Test Quadrants

Quadrant	Test Types	Insurance Focus
Q1: Tech-facing /Support	Unit tests, component tests	Premium calculation, underwriting engine, reinsurance algorithms
Q2: Business-facing/Support	Functional tests, E2E, UAT	Policy issuance, claims, renewal payment, policy servicing
Q3: Business-facing/Critique	Exploratory tests, scenario tests	Exception flows, edge cases, regulatory scenarios
Q4: Tech-facing /Critique	Performance tests, security tests	Peak season load testing, data security compliance

3. Test Case Design Standards

3.1 Case Template

```
# TC-[Module]-[ID]: [Title]
## Basic Info
- Related Requirement: [BRD/TRD ID]
- Test Type: [Functional/Accounting/Actuarial/Integration/Performance]
```

```

- Priority: [P0/P1/P2/P3]
- Automatable: [Yes/No]
## Prerequisites
1. [Condition 1]
## Steps
1. [Step 1]
2. [Step 2]
## Expected Results
1. [Result 1]
2. [Result 2]
## Test Data
| Field | Value | Description |
| --- | --- | --- |
| [Field] | [Value] | [Description] |
## Cleanup
[Data cleanup after test]

```

3.2 Case Design Principles

1. **Equivalence Partitioning** — Cover positive, negative, and boundary values for each business rule
2. **Boundary Value Analysis** — Numeric fields like premium calculation, rate ranges, deductibles
3. **Decision Table** — Multi-condition combinations in underwriting rules (health + occupation + sum assured)
4. **State Transition** — Policy lifecycle (apply > issue > active > claim > terminate)
5. **Exception Scenarios** — Timeout, retry, idempotency, concurrency, data inconsistency
6. **Data-Driven** — Use parameterized data covering multiple products, channels, customer types

4. Accounting Test Standards

4.1 Accounting Test Scope

Accounting testing is the most critical part of insurance core system testing, covering:

Accounting Type	Test Points
Premium Collection	Initial premium, renewal premium, loading/reduction, free-look cancellation
Claims Payment	Claim payout, maturity benefit, survival benefit, dividend payment
Commission Settlement	First-year commission, renewal commission, team management bonus
Reinsurance Settlement	Treaty reinsurance ceded, facultative reinsurance, reinsurance recovery
Reserves	Unearned premium reserve, outstanding claims reserve
Tax Treatment	VAT on premiums, stamp duty, withholding tax

4.2 Verification Methods

Method 1: T-Account Verification

[T-Account – Premium Collection]

Debit (Premium Receivable) Credit (Premium Income)

```

+-----+ +-----+
| 1000.00 | | 1000.00 |
+-----+ +-----+

```

Method 2: Accounting Entry Verification

Transaction: Collect initial premium 1000

Debit: Bank Deposit 1000.00
 Credit: Premium Income – Initial 1000.00
 Transaction: Pay commission 200
 Debit: Commission Expense 200.00
 Credit: Bank Deposit 200.00

Method 3: Balance Verification

Opening Balance + Current Income – Current Expense = Closing Balance
 $0 + 1000 - 200 = 800$

4.3 Accounting Test Checklist

- ☐ Single transaction accounting entries are correct
- ☐ Concurrent transaction accounting summaries are correct
- ☐ Multi-currency exchange rate is correct
- ☐ Reversal/red-ink entries are correct
- ☐ Cross-day, cross-month, cross-year accounting carry-forwards are correct
- ☐ Reconciliation with financial system matches

5. Actuarial Interface Verification

5.1 Verification Scope

Actuarial Interface	Verification Content
Rate calculation	Age/gender/sum assured/payment term > premium output
Cash value	Cash value per policy year, post-reduction cash value
Reserve assessment	Reserve amounts under different valuation standards
Profit testing	Embedded value, new business value
Reinsurance allocation	Retention, ceded amount, reinsurance premium

5.2 Verification Method

```
## Test Case: Rate Calculation API
### Input
- Product Code: PROD-001 (Whole Life)
- Age: 35
- Gender: M
- Sum Assured: 1,000,000
- Payment Term: 20 years
- Payment Frequency: Annual
### Expected Output
- Annual Premium: 12,350.00 (matching actuarial pricing model)
- Monthly Premium: 1,050.00
### Validation Rules
- Difference from actuarial model: < 0.01
- Monthly x 12 > Annual (monthly has surcharge)
- Premium scales linearly with sum assured
```

5.3 Acceptance Criteria

- Rate calculation deviation from actuarial model: less than 0.01
- Cash value table: 100% field match with actuarial model
- Reserve deviation under different standards: within acceptable range

- Actuarial API response time: within SLA (P99 < 2s)

6. Test Execution & Reporting

6.1 Test Phases

Phase	Executor	Entry Criteria	Exit Criteria
Unit Test	Developer	Code complete	UT coverage > 80%
Integration Test	Developer + QA	UT passed	Core interface tests passed
System Test	QA	IT passed	All test pass rate > 95%
UAT	Business	System test passed	Key business scenarios verified
Regression Test	QA	Version change	Core cases 100% passed

6.2 Test Report Template

```
# Test Report: [Project] v[Version]
## Summary
- Period: YYYY-MM-DD ~ YYYY-MM-DD
- Total Cases: [X]
- Passed: [X] | Failed: [X] | Blocked: [X] | Skipped: [X]
- Pass Rate: [X]%
## Defect Statistics
- Total Defects: [X]
- P0: [X] | P1: [X] | P2: [X] | P3: [X]
- Fixed: [X] | Pending: [X]
- Remaining Defect Impact: [Description]
## Test Coverage
- Functional Coverage: [X]%
- Code Coverage: [X]%
- Interface Coverage: [X]%
## Key Findings
1. [Key Finding 1]
2. [Key Finding 2]
## Release Recommendation
[Recommendation: Pass / Conditional Pass / Fail]
```

7. Appendix

7.1 Reference Documents

- ISTQB Foundation Certification
- *Requirements Specification Standard*
- *Performance Testing Standard*
- *Release Checklist Standard*

7.2 Recommended Tools

Purpose	Recommended Tool	Alternative
Unit Testing	JUnit / pytest / go test	TestNG
Mocking	Mockito / MockK / unittest.mock	WireMock
DB Testing	Testcontainers / H2	DbUnit
API Testing	RestAssured / Postman / Bruno	JMeter

Purpose	Recommended Tool	Alternative
E2E Testing	Selenium / Cypress	Playwright
Accounting Testing	In-house verification framework	Excel validation
Coverage	JaCoCo / coverage.py / gocov	---
Test Management	Xray / TestRail	Allure